

SDE Placement Roadmap

C++ · DSA · Competitive Programming · Off-Campus Product Companies

MCA → Google · Amazon · Microsoft · Adobe · Flipkart

**6–12
Months**

Total duration

**400–700
Problems**

DSA target

**7.5+
CGPA**

Stay eligible

**Weekly
Contests**

CF + LC

**1
Internship**

Apply Phase 3

5-PHASE ROADMAP — START TO END

Phase 1

C++ Mastery + Foundation

Month 1

~30

problems

- C++ syntax, STL (vectors, maps, sets, pairs, stack, queue)
- Pointers, references, OOP basics
- Easy Array + String problems only — no trees/graphs yet
- Register on LeetCode + Codeforces. Don't contest yet.
- Set up GitHub. Create LinkedIn profile. Start daily commit habit.
- CGPA: maintain 7.5+ in MCA semesters from Day 1 — non-negotiable
- Resource: Apna College C++ full playlist (Hindi)

Sample problems

- Two Sum (LC #1)
- Reverse Array
- Palindrome Check
- Factorial / Fibonacci via recursion
- Max sub-array sum

Phase 2

Core DSA with Apna College

Month 2–3

~100

problems

- Arrays in depth, Strings, Recursion basics
- Linked List (singly + doubly), Stack, Queue
- Watch Apna College video → immediately solve 3 problems on that topic
- Start Codeforces Div 4 / Div 3 contests on weekends — target A & B only
- Maintain notebook: approach + complexity for every problem
- Start building your 1 project — Library System or Task Manager in C++
- Apply for internships on Internshala + LinkedIn — don't wait to feel ready
- Resource: Apna College DSA playlist (Hindi)

Sample problems

- Merge Sorted Arrays
- Valid Parentheses (LC #20)
- Reverse Linked List (LC #206)
- Subsets via recursion (LC #78)
- Implement Stack using arrays

Phase 3

Striver A2Z Sheet + CS Subjects + Aptitude

Month 4–6

~200

problems

- Binary Search + Search on Answer pattern
- Trees: Binary Tree + BST full series
- Graphs: BFS, DFS, Dijkstra, Topo Sort
- Heaps + Priority Queue, Greedy Algorithms
- DP basics (start Aditya Verma DP playlist)
- ADD: OS + DBMS — Gate Smashers playlist — 1 Pomodoro/day
- ADD: Aptitude — IndiaBIX + GFG Aptitude — 20 min/day
- ADD: English communication — practice explaining solutions aloud
- Upsolve every problem you couldn't solve in contest
- Resources: Striver A2Z Sheet · Aditya Verma DP · Abdul Bari Greedy

Sample problems

- Binary Search on Answer (LC #875)
- Level Order Traversal (LC #102)
- Number of Islands (LC #200)
- Course Schedule (LC #207)
- Longest Increasing Subsequence (LC #300)

Phase 4

LeetCode Med/Hard + LLD + Hackathons

Month 7–9

~200
problems

- Full DP (all patterns): trees, grids, partition, bitmask
- Advanced Graphs: Bellman-Ford, Floyd-Warshall, DSU
- Trie, Segment Tree, Bit Manipulation, Sliding Window
- LeetCode by company tag: Amazon top 50, Google top 50
- Target Codeforces rating: 1200 → 1400+
- ADD: LLD — Parking Lot, ATM, Library System (github.com/ashishps1/awesome-llid)
- ADD: OOPs C++ — Saurabh Shukla Sir playlist
- ADD: Hackathons — apply for SIH, HackWithInfy, Goldman Sachs Hackathon
- ADD: Open Source — find 1 project and raise your first PR
- Start mock interviews — 45 min per problem max
- Resources: Striver DP series · NeetCode 150 · Codeforces EDU

Sample problems

- Trapping Rain Water (LC #42)
- Word Break II (LC #140)
- Edit Distance (LC #72)
- Alien Dictionary (LC #269)
- Serialize & Deserialize Binary Tree (LC #297)

Phase 5

Interview Prep + Placement Sprint

Month 10–12

~100+
problems

- Revise ALL topics — 2–3 problems per topic as revision
- Company-specific lists: Amazon top 50, Google top 50, Adobe top 30
- Mock interviews: Pramp, InterviewBit, peers (8–10 mocks minimum)
- Revise CS subjects: OS, DBMS, CN — GFG interview Q lists
- Practice spoken English explanations — record yourself
- Apply aggressively: LinkedIn, Instahyre, Cutshort, company career pages
- Build referral network: connect with SDEs → 2–3 conversations → ask for referral
- Company research: 30 min night before every interview — tech stack + why them
- Revise aptitude: 1 mock test/week on IndiaBIX
- Resource: Striver SDE Sheet (191 problems) for fast revision

Sample problems

- All revision
- Mock tests on InterviewBit
- LC company-tagged mediums
- Pramp peer interviews
- Apply aggressively — LinkedIn + referrals

WHO TO WATCH FOR WHAT — TOPIC-WISE SOURCE MAP

Step 1 = Learn here first

Step 2 = Go deeper

Step 3 = Practice here

Alt = Alternative / supplement

C++ Language + STL

Phase 1

- S1** Apna College — Shradha Khapra's full C++ playlist. Syntax, pointers, OOP, STL in Hindi.
- S2** cppreference.com — bookmark for STL lookup during problem solving.
- S3** LeetCode: first 20 easy problems in C++.

Arrays + Strings

Phase 1–2

- S1** Apna College — Arrays + Strings playlist. Builds intuition from scratch.
- S2** Striver A2Z Sheet — Arrays section (40+ problems in sequence).
- S3** LeetCode: easy → medium. Sort by 'Arrays' tag. Target 25–30 problems.

Recursion + Backtracking

Phase 2

- S1** Aditya Verma — Recursion playlist. Teaches hypothesis → induction → base case method. Best in class.
- S2** Striver A2Z — Recursion + Backtracking section for harder problems.
- S3** LeetCode: Subsets (#78), Permutations (#46), N-Queens (#51), Sudoku Solver (#37).
- *Do NOT skip Aditya Verma here. This is the only way recursion will truly click.*

Linked List

Phase 2

- S1** Apna College — Linked List playlist. Singly + doubly + circular with visuals.
- S2** Striver — Linked List series on YouTube. Covers LRU Cache, Merge K Sorted.
- S3** LeetCode: Reverse LL (#206), Detect Cycle (#141), Merge Two Sorted (#21).

Stack + Queue

Phase 2

- S1** Aditya Verma — Stack playlist. Best for pattern problems: NGE, Stock Span, Histogram.
- S2** Striver A2Z — Stack + Queue section. Monotonic stack problems.
- S3** LeetCode: Valid Parentheses (#20), Min Stack (#155), Largest Rectangle (#84).
- *Aditya Verma's stack playlist is uniquely good — don't substitute it.*

Binary Search

Phase 3

- S1** Striver — Binary Search series. Teaches 'search on answer' pattern for hard problems.
- S2** Aditya Verma — Binary Search playlist for additional pattern clarity.
- S3** LeetCode: Search Rotated Array (#33), Koko Bananas (#875), Median Two Arrays (#4).

Sliding Window + Two Pointer

Phase 3

- S1** Aditya Verma — Sliding Window playlist. Dedicated series. Fixed + variable window templates.
- S3** LeetCode: Longest Substring (#3), Min Window Substring (#76), Max Consecutive Ones III (#1004).
- *This pattern appears in almost every Amazon OA.*

Trees (BT + BST)

Phase 3

- S1** Striver — 'Free ka Tree' series on YouTube. 50+ videos, every tree pattern covered. Most complete free resource.
- S2** Apna College Trees playlist for initial introduction before Striver.
- S3** LeetCode: Level Order (#102), Diameter (#543), LCA (#236), Serialize/Deserialize (#297).

Heap + Priority Queue

Phase 3

- S1** Aditya Verma — Heap playlist. Builds heap from scratch + all classic problems.
- S2** Striver A2Z — Heap section for C++ STL priority_queue usage.
- S3** LeetCode: Top K Frequent (#347), K Closest Points (#973), Median from Stream (#295).

Graphs (BFS + DFS)

Phase 3

- S1** Striver — Graph Series on YouTube. BFS, DFS, Dijkstra, Bellman-Ford, Topo Sort, DSU. Most complete.
- S2** Abdul Bari — watch for theoretical foundation: Dijkstra proof, MST concepts.
- S3** LeetCode + Codeforces: Number of Islands (#200), Course Schedule (#207), Codeforces Div 3 graph Qs.

Greedy Algorithms

Phase 3

- S1** Abdul Bari — Greedy playlist. Clearest explanation of when greedy works vs fails. Theory-first.
- S2** Striver A2Z — Greedy section for pattern-based problems after Abdul Bari.
- S3** Codeforces Div 3 — B and C problems are often greedy. Best pressure practice.

Dynamic Programming

Phase 3–4

- S1** Aditya Verma — DP playlist FIRST. Recursion → Memoization → Tabulation → Space Optimization. Covers Knapsack, LCS, MCM, DP on Strings.
- S2** Striver — DP series (50+ videos). Covers DP on Trees, Grids, Partition DP, Bitmask DP.
- S3** LeetCode: LIS (#300), Edit Distance (#72), Coin Change (#322), Burst Balloons (#312).
- *ALWAYS start with Aditya Verma for DP. Starting with Striver directly will confuse you.*

Trie

Phase 4

S1 Striver — Trie series. Implement from scratch in C++ then solve problems. XOR Trie covered.

S3 LeetCode: Implement Trie (#208), Word Search II (#212), Max XOR (#421).

Bit Manipulation

Phase 4

S1 Striver A2Z — Bit Manipulation section. AND/OR/XOR tricks, counting bits, power of two.

S2 GFG Bit Manipulation tricks article — bookmark for quick contest reference.

S3 LeetCode: Single Number (#136), Counting Bits (#338), Sum Without + (#371).

CS SUBJECTS — SOURCE MAP (1 Pomodoro/day from Phase 3)

Operating Systems

Primary: Gate Smashers — OS playlist (YouTube, Hindi). Processes, threads, scheduling, deadlock, memory mgmt, paging.

Supplement: GFG OS Interview Questions — use as revision checklist before interviews only.

Key topics: Topics: Process vs Thread · CPU Scheduling · Deadlock · Paging · Semaphore · Virtual Memory

DBMS + SQL

Primary: Gate Smashers — DBMS playlist (YouTube, Hindi). Normalization, ACID, ER diagrams, indexing, transactions.

Supplement: LeetCode SQL section (free) — do top 20 most liked SQL problems. Also SQLZoo for query practice.

Key topics: Topics: ACID Properties · Normalization 1NF-BCNF · SQL Joins · Indexes & B-Trees · RDBMS vs NoSQL

Computer Networks

Primary: Neso Academy — CN playlist (YouTube, English). OSI model, TCP/IP, DNS, HTTP/HTTPS, TCP handshake.

Supplement: Gate Smashers CN playlist (Hindi) — if you prefer Hindi over Neso Academy.

Key topics: Topics: OSI 7 Layers · TCP vs UDP · TCP 3-Way Handshake · DNS · HTTP vs HTTPS · What happens when you type a URL

OOPs in C++

Primary: Saurabh Shukla Sir — C++ OOPs playlist (YouTube, Hindi). All 4 pillars, virtual functions, vtable, abstract classes.

Supplement: Striver — OOPs playlist for interview-focused Q&A; style revision.

Key topics: Topics: Encapsulation · Abstraction · Inheritance · Polymorphism · Virtual Functions · SOLID Principles

LLD + Design Patterns

Primary: github.com/ashishps1/awesome-low-level-design — free, curated. Parking Lot, ATM, Library, Elevator in C++.

Supplement: Refactoring.Guru — Singleton, Factory, Observer, Strategy, Decorator with C++ examples.

Key topics: Practice: Parking Lot → Library System → ATM — implement each from scratch without looking at solution

PLATFORMS — LEARN · PRACTICE · SHOWCASE · APPLY

LEARN (YouTube)

Apna College	C++ full course + Core DSA (Arrays, LL, Stack, Queue, Trees)	Phase 1–2 · Hindi · Free
Striver / takeUforward	A2Z DSA, Graphs, Trees, DP series. Ex-Google, Ex-Amazon.	Phase 2–4 · English · Free
Aditya Verma	DP, Recursion, Stack, Heap, Sliding Window — uniquely deep playlists.	Phase 2–4 · Hindi · Free
Abdul Bari	Greedy, Sorting, Graph theory — mathematical rigour.	Phase 3 · English · Free
NeetCode	Short pattern-focused LC walkthroughs. neetcode.io roadmap.	Phase 4–5 · English · Free
Gate Smashers	OS + DBMS playlists — best free CS subject resource.	Phase 3 · Hindi · Free
Neso Academy	Computer Networks playlist — methodical, clear.	Phase 3 · English · Free
Saurabh Shukla Sir	C++ OOPs — all 4 pillars, virtual functions, vtable.	Phase 2–3 · Hindi · Free

SHEETS & NOTES

Striver A2Z Sheet	455 problems, topic-by-topic. takeuforward.org — your Phase 3 bible.	Phase 3 · Free
Striver SDE Sheet	191 problems. Fast revision before interviews.	Phase 5 · Free
NeetCode 150	neetcode.io — 150 curated LC problems by pattern.	Phase 4 · Free
GFG Interview Qs	OS + DBMS + CN questions compiled from 2000+ real interviews.	Phase 5 · Free
awesome-ll	github.com/ashishps1/awesome-low-level-design — LLD problems + C++ code.	Phase 4 · Free

PRACTICE PLATFORMS

LeetCode	Interview gold standard. Company-tagged lists, weekly contests.	Phase 2–5 · Free tier
Codeforces	Best for CP. Div 4 → Div 3 → Div 2. Target: 1200 → 1400+ rating.	Phase 2–5 · Free
CodeChef	Indian platform, Starters contests. India-friendly timings.	Phase 2–3 · Free
InterviewBit	Structured topic practice + timed company mock tests.	Phase 4–5 · Free tier
Pramp	Free peer-to-peer mock interviews. Essential for communication practice.	Phase 5 · Free
LeetCode SQL	70+ free SQL problems. Do top 20 most liked.	Phase 3 · Free

SHOWCASE & APPLY

GitHub	Pin 3–4 best repos. Daily commits. DSA solutions by topic. Profile in resume bio.	Start Phase 1
LeetCode Profile	Problem count + contest rating visible. Put URL on resume. 300+ = strong signal.	On your resume
Codeforces Handle	Rating on resume. 1200 (Pupil) = decent. 1400 (Specialist) = impressive.	On your resume
LinkedIn	Post weekly. Connect with SDEs. Follow Adobe/Flipkart/PhonePe for off-campus forms.	Start Phase 3
Jake's Resume	overleaf.com — best ATS-friendly 1-page LaTeX template. Used by placed candidates.	Phase 4
Instahyre/Cutshort	Product-company focused portals. Razorpay, Groww, CRED, Zepto hire here.	Phase 5

DAILY POMODORO SCHEDULE (40 min work / 10 min break)

MON-FRI (College Day)

6:00–6:40	Learn	Apna College / Striver video — 1 topic only. Watch at 1.25x. Trace code by hand. No passive watching.
6:50–7:30	Solve	1–2 easy problems on today's topic. No hints for 20 min. Write approach first, then code.
2:00–2:40	Solve	2 fresh problems on the same topic. 30 min solo attempt each. Read editorial if stuck — understand then rewrite.
2:50–3:30	Solve	1 problem from yesterday's topic. Spaced repetition — forces recall and reinforces patterns.
3:40–4:20	Review	Revisit today's solutions. Can you write faster? Write time + space complexity for every solution.
9:30–10:10	Notes	Notebook entry: topic learned + 3 key patterns + 1 mistake made today. This becomes your interview cheat sheet.
10:20–11:00	Solve	1 LeetCode easy — free topic, no pressure. Builds confidence and problem count.

SAT-SUN (Full Grind)

~5 problems · 4.5 hrs

7:00–8:20	Learn	2 Apna College / Striver videos. Cover a full subtopic. Take notes. Do NOT skip to next topic.
8:30–10:10	Solve	3–4 problems on today's topic. All solo. Check after each: can solution be improved?
2:00–3:40	Contest	Codeforces Div 4 / virtual Div 3 contest. Saturday: attempt live. Target A + B problems only.
3:50–4:30	Upsolve	Every problem not solved: read editorial → understand fully → code from scratch. This is the REAL learning.
9:00–9:40	Review	Weekly review: how many problems solved? Which topics felt weak? What to fix next week?

GOLDEN RULES — NEVER BREAK THESE

1	Attempt 30 min solo before looking at any hint or solution.	1	Don't watch tutorials without solving problems after. Tutorial hell is real.
2	Never copy-paste a solution — always understand and rewrite.	2	Never skip a weekly contest — even if you solve only 1 problem, show up.
3	Upsolve every contest problem you couldn't solve. That's the real training.	3	Speak your approach out loud when solving from Phase 3 onwards.
4	Write time + space complexity for every problem you solve.	4	One project built well beats five projects half-built.
5	Stick to C++ — do NOT switch languages at any point.	5	Post on LinkedIn weekly — recruiter visibility compounds over time.
6	Daily practice beats weekend marathon. Never take 2 days off in a row.	6	In interviews: explain your thinking, not just your code.

BEYOND DSA — COMPLETE SKILL CHECKLIST

DSA + Competitive Programming

Phase 1–4 400–700 problems · LeetCode + Codeforces · Striver A2Z Sheet

OOPs in C++

Phase 2–3 4 pillars · Virtual functions · vtable · SOLID principles · Design patterns

Operating Systems

Phase 3 Processes · Scheduling · Deadlock · Memory Mgmt · Semaphores · Paging

DBMS + SQL

Phase 3 ACID · Normalization · SQL Joins · Indexing · Transactions · NoSQL basics

Computer Networks	Phase 3	OSI Model · TCP/UDP · DNS · HTTP/HTTPS · TCP Handshake · URL flow
Low Level Design (LLD)	Phase 4	Parking Lot · Library System · ATM · Elevator · Singleton · Factory Pattern
1 Solid Project on GitHub	Phase 2–3	Explainable in 2 min · Clean README · Full code you wrote yourself
Resume — Jake's Template	Phase 4	1 page · ATS-friendly · LeetCode + CF links · No photos · No colors
LinkedIn + Referral Network	Phase 3+	Post weekly · Connect with SDEs · Follow company pages for off-campus drives
Behavioral + STAR Stories	Phase 4–5	3 stories ready: challenge faced · failure + learning · project you're proud of
Mock Interviews	Phase 5	8–10 mocks minimum · Pramp · InterviewBit · Peer practice · Time yourself
CGPA — Maintain minimum 7+	Always	Flipkart cutoff 6.5 · Goldman 7.0 · Most product companies 7–7.5 · Never neglect academics
Aptitude + Reasoning	Phase 3–5	Quant · Logical · Verbal English · IndiaBIX · GFG Aptitude · 20–30 min/day
English Communication	Phase 3+	Speak solutions aloud · Practice interviews in English · Verbal OA sections
Internship	Phase 3–4	Even 2-month unpaid startup internship changes resume perception · Internshala · AngelList
Maths — Number Theory + Combinatorics	Phase 3–4	Modular arithmetic · Prime sieve · GCD/LCM · Permutations · Probability
Company Research	Phase 5	30 min before every interview · Tech stack · Engineering blog · Recent news · Why this company
Hackathons — company-sponsored	Phase 4–5	Smart India Hackathon · HackWithInfy · Goldman Sachs · JP Morgan Code for Good
Open Source Contributions	Phase 4–5	2–3 small meaningful contributions · Bug fix or feature · Real project on GitHub

THINGS MOST PEOPLE MISS — BUT CANNOT AFFORD TO

CGPA — the silent eligibility gate

- Flipkart minimum: 6.5 CGPA | Goldman Sachs: 7.0 CGPA
- Most product companies filter at 7.0–7.5 before any human sees your resume
- Cutoff applies to 10th, 12th, AND graduation — not just final semester
- A 600-problem LeetCode profile will NOT save you if CGPA is below cutoff
- Target: maintain 7.5+ throughout your MCA — consistent, not last-minute

Aptitude + Reasoning — eliminates most people in round 1

- Most OAs have: Quantitative Aptitude + Logical Reasoning + Verbal English
- Adobe, D.E. Shaw, and many mid-size product OAs are 40–50% aptitude
- Strong DSA with weak aptitude = eliminated before the coding round
- Topics: Number System, Time-Work, Probability, Permutations, Seating Arrangements
- Preparation: IndiaBIX + GFG Aptitude · 20–30 min/day from Phase 3 onwards

English Communication — the skill that loses offers after solving the problem

- Verbal ability is tested in OAs: comprehension, grammar, sentence arrangement
- In interviews: interviewers score you on HOW you explain, not just what you code
- Practice speaking your approach aloud in English from Phase 3 — every session
- Format to practice: 'I am thinking of using a HashMap here because lookup is $O(1)$ '
- If you fumble in English during interviews, you lose offers even after correct solutions

Internship — single biggest resume differentiator from tier 3 college

- Even a 2-month unpaid startup internship changes how your resume is read
- It signals: you have shipped real code, worked in a team, used Git professionally
- Apply from Phase 3 onwards — Internshala, LinkedIn, AngelList, Wellfound
- Do NOT wait until you feel 'ready' — apply now, learn on the job
- One internship entry on resume outweighs 100 extra LeetCode problems for shortlisting

Mathematics — number theory + combinatorics appear more than expected

- Modular arithmetic, Prime Sieve (Sieve of Eratosthenes), GCD/LCM — standard OA topics
- Permutations and Combinations, Probability — appear in Codeforces Div 2 C/D problems
- These come naturally through competitive programming — no separate textbook needed
- Just be aware: CF Div 2 C problems often require combinatorics intuition
- Resource: GFG Mathematics for CP + Striver A2Z Math section

Company research — 30 minutes the night before every interview

- Before every interview: research what the company builds, their tech stack, recent news
- Read their engineering blog — Flipkart Tech, Amazon Science, Adobe Tech Blog
- Prepare a specific answer for 'Why do you want to join us?' — not a generic one
- Interviewers immediately notice when a candidate knows nothing about their company
- This costs zero preparation time during your study phase — just do it the night before

Referrals — the highest-value off-campus channel that most students ignore

- 30% of hiring at product companies happens through employee referrals
- A referral guarantees your resume reaches a human instead of an ATS filter
- How to get one: connect with SDEs on LinkedIn → have 2–3 genuine conversations → ask
- Never cold-ask for a referral — build the relationship first over weeks
- Companies with referral portals: Amazon, Microsoft, Flipkart, Adobe, Paytm

Hackathons — worth your time only if they are company-sponsored hiring pipelines

- Smart India Hackathon (SIH) — national recognition, resume value
- HackWithInfy — direct shortlisting pipeline for Infosys product roles
- Goldman Sachs India Hackathon — top finalists get direct interview opportunity (PPI)
- JP Morgan Code for Good — used as talent acquisition event
- Random online Devfolio hackathons with no company involved: skip them

Open source — 2–3 meaningful contributions is enough. Don't obsess.

- Recruiters DO check GitHub — green commit graph signals discipline and consistency
- You do NOT need to contribute to React or Linux — a mid-sized project is enough
- Find a project in your stack → fix a real bug or add a missing feature → raise a PR
- Even improving documentation with code examples counts as a contribution
- Start 2–3 months before placement season — one confirmed merged PR is enough to mention

COMPLETE PRIORITY ORDER — WHAT TO DO WHEN

Priority	Skill / Activity	When to Start	Why It Matters
1	DSA + Competitive Programming	Phase 1 — Day 1	Clears every OA coding round
2	CGPA — maintain 7.5+	Always — right now	Eligibility cutoff. No CGPA = no shortlist
3	CS Subjects (OS, DBMS, CN, OOPs)	Phase 3	Asked in same round as DSA at MS/Adobe
4	Aptitude + Verbal English	Phase 3	Eliminates 60% of candidates in OA round 1
5	1 Solid Project on GitHub	Phase 2–3	Gets you the shortlist call
6	Internship	Phase 3–4	Biggest resume differentiator from tier 3
7	Resume + LinkedIn	Phase 3 onwards	LinkedIn = off-campus hiring channel
8	English Communication	Phase 3 onwards	Loses offers even after correct solutions
9	LLD + Behavioral + STAR Stories	Phase 4–5	Survives the interview, gets the offer
10	Company Research + Referrals	Phase 5 / Pre-interview	Highest ROI per hour activity in Phase 5
11	Hackathons (company-sponsored only)	Phase 4–5	SIH, HackWithInfy = direct PPI pipelines
12	Open Source (2–3 contributions)	Phase 4–5	GitHub signal of discipline to recruiters

DSA clears the OA	CGPA gets you eligible	Aptitude survives round 1	OOPs survive interview	internship gets shortlisted	referrals get the call	Consistency gets the offer
-------------------	------------------------	---------------------------	------------------------	-----------------------------	------------------------	----------------------------